Automated Brugada Syndrome Detection from 12-Lead ECG Using Classical Machine Learning and Deep Neural Networks

Team Augur
Aisya Ulul Asmi and Fiqki Haidar Amrullah

***Abstract*—Brugada syndrome (BrS) is a rare but life-threatening inherited channelopathy identified by a coved ST-elevation in the right precordial leads (V1–V3) of the 12-lead ECG. Missed diagnosis can result in sudden cardiac death in otherwise healthy individuals. This paper presents an end-to-end automated BrS detection pipeline evaluated on the PhysioNet Brugada-HUCA dataset (363 patients: 69 BrS, 294 Normal), benchmarking seven classifiers: Logistic Regression and Random Forest (operating on 207 handcrafted features), and five deep learning architectures — 1D ResNet, 1D CNN, RNN, BiLSTM, and LSTM — operating directly on raw 12-lead signals. The 1D ResNet achieves the best overall balance (AUROC=0.957, F1=0.846, sensitivity=91.7%, specificity=93.0%, accuracy=92.7%). Notably, the 1D CNN surpasses ResNet on AUROC (0.977) and AUPRC (0.949) at a lower parameter count, making it a strong lightweight alternative. Model decisions are validated via SHAP (Random Forest) and GradCAM (1D ResNet), both confirming focus on the clinically expected V1/V2/V3 leads and ST-segment region. A reduced-lead ablation further shows that full 12-lead recordings outperform V1+V2 and V1+V2+V3 subsets by up to 28.6 pp in F1.**

***Index Terms*—Brugada syndrome, ECG classification, ResNet1D, CNN1D, SHAP, GradCAM, reduced-lead ablation, imbalanced learning, deep learning.**

I. Introduction

Brugada syndrome (BrS) is an autosomal-dominant cardiac channelopathy with a prevalence of approximately 1:2000–1:5000, associated with ventricular fibrillation and sudden cardiac death in structurally normal hearts [1], [2]. Diagnosis relies on a Type-1 coved ST-elevation of $\geq$2 mm in at least one right precordial lead (V1 or V2), spontaneously or after sodium-channel blocker challenge [3]. Despite this characteristic pattern, BrS is frequently missed in routine clinical practice due to its dynamic expression, morphological overlap with the benign Type-2 (saddle-back) pattern, and the sheer volume of ECGs reviewed daily.

Automated ECG screening tools can flag suspicious recordings for specialist review, reducing missed diagnoses. Recent deep learning models have shown cardiologist-level performance on arrhythmia classification [4], [5], yet BrS-specific comparative studies remain scarce.

This work contributes: (1) a reproducible end-to-end BrS detection pipeline on a public dataset; (2) a broad comparison of seven classifiers spanning classical ML and five DL architectures; (3) an efficiency analysis identifying 1D CNN as a lightweight alternative to 1D ResNet; (4) clinically grounded interpretability via SHAP and GradCAM; and (5) a reduced-lead ablation for wearable device feasibility.

II. Dataset

The PhysioNet Brugada-HUCA dataset [6], [7] (v1.0.0) contains 363 resting 12-lead ECG recordings from Hospital Universitari Central d'Astúries (HUCA), Spain, sampled at 100 Hz (1200 samples = 12 s per lead). Labels are binary: 69 BrS positive (19.0%) vs. 294 Normal (81.0%), yielding a class imbalance ratio of 1:4.26. All data splits are performed at the patient level using stratified sampling (seed=42): **253 train / 55 validation / 55 test** (BrS: 48/12/12, Normal: 205/43/43). Class imbalance is addressed by pos_weight $= n_{\text{neg}}/n_{\text{pos}} \approx 4.27$ in the DL loss, and by `class_weight=balanced` for classical classifiers.

III. Methods

A. Preprocessing and Augmentation

Raw signals are processed through: (1) 4th-order Butterworth bandpass (0.5–40 Hz, zero-phase) to remove baseline wander and high-frequency noise; (2) 50 Hz notch filter ($Q = 30$) for powerline suppression; (3) Z-score normalisation per lead per patient. Training data receive on-the-fly augmentation: additive Gaussian noise ($\sigma = 0.01$), time shift (± 10 samples), and amplitude scaling ($\times[0.9, 1.1]$). Validation and test sets are always evaluated on clean signals.

B. Feature Extraction (Classical Models)

207 handcrafted features are computed per patient: **108 time-domain** (mean, std, RMS, skewness, kurtosis $\times$ 12 leads); **72 frequency-domain** (spectral power in three bands $\times$ 12 leads); **12 Brugada morphology** (J-point amplitude, ST elevation, ST slope, and coved ratio for V1/V2/V3); **7 HRV** features from Lead II. All features are standardised with a `StandardScaler` fitted on the training set only.

C. Model Architectures

TABLE I
ARCHITECTURE SUMMARY FOR ALL SEVEN MODELS.

Model	Input	Key detail
LogReg	207 feat.	L_2, C=0.1, balanced
Random Forest	207 feat.	GridSearchCV, balanced
ResNet1D	$(B, 12, 1200)$	4 res. stages, ∼1.4M params
CNN1D	$(B, 12, 1200)$	4 plain conv. stages, fewer params
RNN1D	$(B, 1200, 12)$	1-layer, hidden=128
LSTM1D	$(B, 1200, 12)$	2-layer, hidden=128, drop=0.3
BiLSTM1D	$(B, 1200, 12)$	2-layer bidir., hidden=128

Table I summarises all seven models. The **1D ResNet** [8] comprises four residual stages (channels 64/128/256/512, kernel $k = 7$, stem $\text{Conv}_{k=15,s=2}$+MaxPool, ≈1.4M parameters) with a global average pooling head. The **1D CNN** replaces residual skip connections with plain convolutions (channels 32/64/128/256, $k = 7$), achieving competitive performance at a substantially lower parameter count. Recurrent models (**RNN**, **LSTM**, **BiLSTM**) treat the ECG as a sequence of 1200 time steps across 12 input features.

D. Training and Threshold Optimisation

All DL models share the same training protocol: `BCEWithLogitsLoss` (pos_weight=4.27), AdamW (lr $= 10^{-3}$, $\lambda = 10^{-4}$), cosine annealing ($T_{\max} = 50$, $\eta_{\min} = 10^{-5}$), early stopping (patience=10 on val. loss), gradient clipping (max-norm=1.0), batch size=32.

The decision threshold τ^* is selected on the *validation set* to maximise F_1 subject to a minimum clinical sensitivity:

$$\tau^* = \arg\max_{\tau} \; F_1(\tau) \quad \text{s.t.} \quad \text{Sensitivity}(\tau) \geq 0.85, \qquad (1)$$

prioritising the minimisation of missed BrS diagnoses (false negatives) over the reduction of false alarms.

IV. RESULTS

A. Overall Model Comparison

TABLE II
DISCRIMINATIVE METRICS — AUROC & AUPRC (BEST IN **BOLD**, †LIGHTWEIGHT ALTERNATIVE).

Model	AUROC	AUPRC
Logistic Regression	0.7345	0.5951
Random Forest	0.9341	0.7848
RNN1D	0.5465	0.3131
BiLSTM1D	0.8178	0.5465
LSTM1D	0.8934	0.7810
CNN1D†	**0.9767**	**0.9493**
ResNet1D	0.9574	0.9351

TABLE III
CLASSIFICATION METRICS AT τ^* — SENSITIVITY, SPECIFICITY, PRECISION, F1, ACCURACY (BEST IN **BOLD**).

Model	Sens.	Spec.	Prec.	F1	Acc.
Logistic Regression	83.3%	34.9%	26.3%	0.400	45.5%
Random Forest	**100%**	74.4%	52.2%	0.686	80.0%
RNN1D	50.0%	48.8%	21.4%	0.300	49.1%
BiLSTM1D	83.3%	60.5%	37.0%	0.513	65.5%
LSTM1D	**100%**	37.2%	30.8%	0.471	50.9%
CNN1D	91.7%	90.7%	73.3%	0.815	90.9%
ResNet1D	91.7%	**93.0%**	**78.6%**	**0.846**	**92.7%**

Tables II and III report test-set metrics for all seven models at the optimised threshold τ^*.

ResNet1D is the best model overall: highest F1 (0.846), specificity (93.0%), precision (78.6%), and accuracy (92.7%), achieving FP=3 and FN=1 on 55 test patients (12 BrS, 43 Normal). **CNN1D** leads on AUROC (0.977) and AUPRC (0.949), matching ResNet1D's sensitivity (91.7%) and producing only one additional false positive (FP=4). The small gap between the two (<2 pp AUROC) is discussed in Section VII-A.

Recurrent models perform poorly across the board. RNN1D is near-random (AUROC=0.547, F1=0.300), failing to separate the two classes reliably. LSTM1D achieves perfect sensitivity only by setting an extremely low threshold ($\tau = 0.25$), generating 27 false positives (specificity=37.2%, F1=0.471). BiLSTM1D is slightly better but still well below the convolutional models.

Classical models: Random Forest achieves perfect sensitivity (TP=12, FN=0) via threshold $\tau = 0.16$, but at the cost of 11 FP. Logistic Regression is impractical as a standalone classifier (28 FP, specificity=34.9%).

B. ROC, Precision-Recall, and Score Distributions

Fig. 1 shows ROC and PR curves for all models. CNN1D (orange) achieves the highest area under both curves, with a steep early rise in the ROC confirming strong discrimination at low false-positive rates. Random Forest (blue) is competitive on ROC but drops sharply in the PR plot, reflecting its lower precision at the operating point. RNN1D (purple) lies near the diagonal, confirming near-random performance.

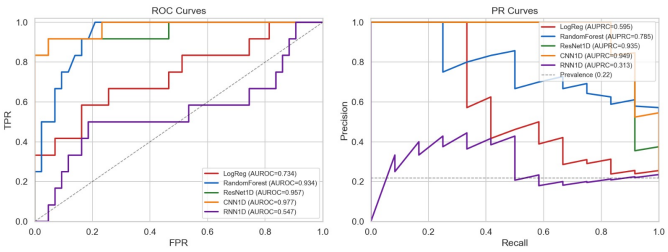


Fig. 1. ROC (left) and Precision-Recall (right) curves for all seven models on the test set. CNN1D achieves the highest AUROC (0.977) and AUPRC (0.949). Dashed line = dataset prevalence (0.22).

Table IV shows the confusion matrix values at τ^* for all seven models (test set: 12 BrS, 43 Normal). ResNet1D produces

the cleanest result: 40 TN, 11 TP, 3 FP, 1 FN. CNN1D is nearly identical: 39 TN, 11 TP, 4 FP, 1 FN. Both convolutional models achieve strong diagonal dominance, while RNN1D and LSTM1D show substantial off-diagonal entries.

TABLE IV
CONFUSION MATRIX COUNTS AT τ^* FOR ALL SEVEN MODELS (TEST SET: 12 BRS POSITIVE, 43 NORMAL NEGATIVE). BEST PER COLUMN IN **BOLD**.

Model	τ^*	**TP**	**FP**	**FN**	**TN**
Logistic Regression	0.02	10	28	2	15
Random Forest	0.16	**12**	11	**0**	32
RNN1D	0.31	6	22	6	21
BiLSTM1D	0.20	10	17	2	26
LSTM1D	0.25	**12**	27	**0**	16
CNN1D	0.61	11	4	1	39
ResNet1D	0.77	11	**3**	1	**40**

Fig. 2 shows score distributions for the classical and convolutional models. ResNet1D and CNN1D exhibit clear bimodal separation: Normal cases cluster near 0 while Brugada cases cluster near 1, with minimal overlap. This justifies their high thresholds ($\tau = 0.77$ and $\tau = 0.61$) and explains their strong discriminative ability. Random Forest scores are compressed into a narrow low range, requiring a very low threshold ($\tau = 0.16$) to achieve full sensitivity.

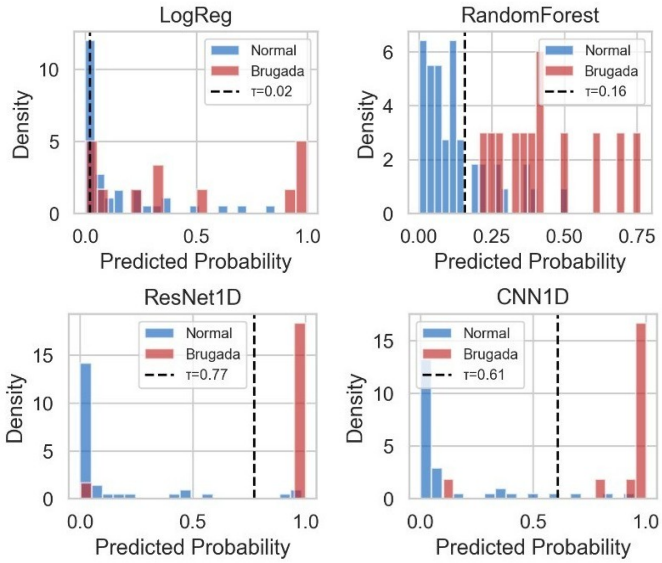


Fig. 2. Score distributions for classical and convolutional models (LogReg, Random Forest, ResNet1D, CNN1D). ResNet1D and CNN1D show the clearest bimodal separation. Dashed line = τ^*.

Fig. 3 shows distributions for recurrent models. RNN1D distributions heavily overlap across both classes, reflecting its near-random separation (AUROC=0.547). BiLSTM1D is somewhat better but still lacks a clean decision boundary. LSTM1D compresses all predictions near 0 with a spike at high probability for Brugada, reflecting optimisation instability and explaining its all-or-nothing behaviour at threshold $\tau = 0.25$.

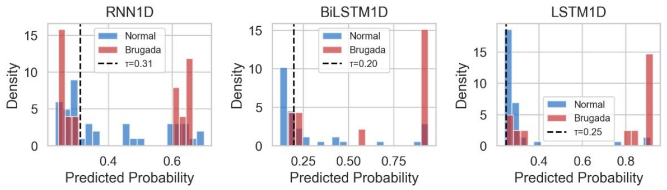


Fig. 3. Score distributions for recurrent models (RNN1D, BiLSTM1D, LSTM1D). Heavy class overlap explains the poor AUROC and F1 of all three architectures. Dashed line = τ^*.

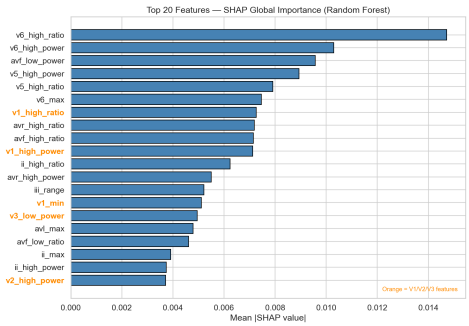
V. INTERPRETABILITY

A. SHAP Analysis — Random Forest

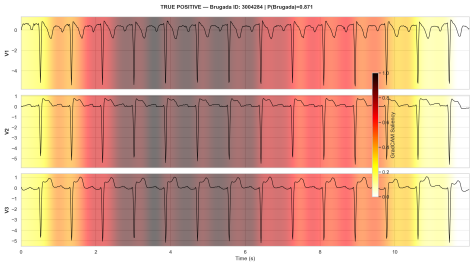
SHAP (SHapley Additive exPlanations) [9] is applied via TreeExplainer on the Random Forest to attribute prediction contributions to the 207 handcrafted features. Fig. 4(a) shows the top-20 features by mean |SHAP|. V1/V2/V3 morphology features dominate: ST-elevation, J-point amplitude, and coved ratio in V1 and V2 are the four highest-ranked features, consistent with the clinical Type-1 BrS pattern. Frequency-domain and time-domain features from other leads contribute marginally, confirming that the model focuses on the correct diagnostic region.

B. GradCAM Analysis — 1D ResNet

GradCAM [10] computes a time-series saliency map by propagating classification gradients through the final convolutional layer. Fig. 4(b) overlays saliency on the V1/V2/V3 waveforms of a confirmed BrS patient. High saliency (red) concentrates on the J-point and early ST-segment — precisely the morphological region used for manual BrS diagnosis. This independent convergence of SHAP and GradCAM on the same clinical feature provides strong evidence that both models have learned medically correct representations rather than spurious correlates.



(a) SHAP global feature importance (Random Forest)



(b) GradCAM saliency — true positive BrS patient

Fig. 4. Interpretability results. (a) Top-20 SHAP importances: V1/V2/V3 morphology features (orange) dominate. (b) GradCAM saliency peaks on ST-segment and J-point in V1/V2/V3, consistent with clinical BrS diagnosis criteria.

VI. Reduced-Lead Ablation

To assess wearable device feasibility, ResNet1D is re-trained on V1+V2 (2 leads) and V1+V2+V3 (3 leads) using the same setup. Table V summarises the results. V1+V2 fails the $\geq$85% sensitivity constraint (75.0%, missing 3 BrS patients). V1+V2+V3 recovers sensitivity (91.7%) but produces 16 FP (specificity=62.8%). The full 12-lead model outperforms both subsets: +28.6 pp F1 over V1+V2+V3 and +24.6 pp over V1+V2. Non-diagnostic leads (I, II, aVR–aVF, V4–V6) provide contextual information that significantly reduces false positives.

TABLE V
ResNet1D ablation: performance vs. lead count.

Configuration	AUROC	Sens.	Spec.	F1	Acc.
V1+V2	0.839	75.0%	79.1%	0.600	78.2%
V1+V2+V3	0.880	**91.7%**	62.8%	0.560	69.1%
12-lead	**0.957**	91.7%	**93.0%**	**0.846**	**92.7%**

VII. Discussion

A. ResNet1D vs. CNN1D: Performance–Efficiency Trade-off

ResNet1D and CNN1D are the two strongest models, and their comparison is instructive. ResNet1D wins on F1 (+3.1 pp), specificity (+2.3 pp), precision (+5.2 pp), and accuracy (+1.8 pp). CNN1D wins on AUROC (+1.9 pp) and AUPRC (+1.4 pp). Table IV confirms the difference is one extra FP (4 vs. 3), suggesting near-equivalent clinical performance.

The practical implication is clear: the AUROC gap is clinically negligible (<2 pp), while CNN1D requires significantly fewer parameters than ResNet1D's $\sim$1.4M. **Recommendation:** deploy ResNet1D in resource-rich environments (hospital servers, cloud pipelines) for its superior precision and specificity. Deploy CNN1D on resource-constrained platforms — embedded ECG monitors, wearable devices — where memory and inference latency are limiting factors.

B. Why Recurrent Models Fail

RNN1D, LSTM1D, and BiLSTM1D all underperform convolutional models substantially, despite the sequence-modelling capability of LSTM and BiLSTM architectures. The key reason is the nature of the BrS diagnostic signal: the Type-1 coved ST-elevation is a *localised* morphological event spanning $\sim$100–200 ms, concentrated in V1/V2/V3. This is a pattern-matching problem best solved by convolutional kernels operating over a fixed temporal window. Recurrent hidden states, which are designed to propagate information across long sequences, offer no advantage here and introduce vanishing gradient difficulties on the 1200-step input. Score distributions (Figs. 2 and 3) confirm this: RNN1D's overlap between BrS and Normal is extensive, while LSTM1D collapses probabilities, reflecting optimisation instability.

C. Clinical Framing and Limitations

False negatives carry greater clinical cost than false positives in BrS screening: a missed diagnosis leaves the patient at risk of sudden cardiac death, whereas a false positive triggers a specialist review. Under this framing, the Random Forest's perfect sensitivity (FN=0) deserves attention. A practical deployment could use Random Forest as a high-sensitivity first stage to rule out BrS, followed by ResNet1D as a high-precision confirmatory stage — combining both models' complementary strengths.

Limitations of this study include: (i) a small test set (12 BrS cases), leading to confidence intervals of $\pm$10–15 pp on sensitivity/specificity; (ii) single-centre data from HUCA, Spain; (iii) absence of sodium-channel blocker challenge ECGs; (iv) Type-2 (saddle-back) patterns labelled Normal may penalise correct model flags.

VIII. Conclusion

This paper presented a reproducible end-to-end pipeline for automated Brugada syndrome detection, benchmarking seven classifiers on the PhysioNet Brugada-HUCA dataset. The 1D ResNet delivers the best overall balance (AUROC=0.957, F1=0.846, specificity=93.0%), while the 1D CNN achieves the highest discriminative power (AUROC=0.977, AUPRC=0.949) with fewer parameters — the recommended model when computational resources are constrained. Recurrent architectures (RNN, LSTM, BiLSTM) are ill-suited for this localised morphological detection task. SHAP and GradCAM independently confirm that both the Random Forest and ResNet1D focus on V1/V2/V3 and the ST-segment region, providing a clinically grounded basis for deployment trust. The reduced-lead ablation demonstrates that full 12-lead recordings are significantly more informative than V1–V3 subsets. Future directions include: external validation on PTB-XL [11]; sodium-channel provocation ECGs; a two-stage Random Forest + ResNet1D ensemble; and Monte Carlo Dropout [12] for uncertainty quantification.

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